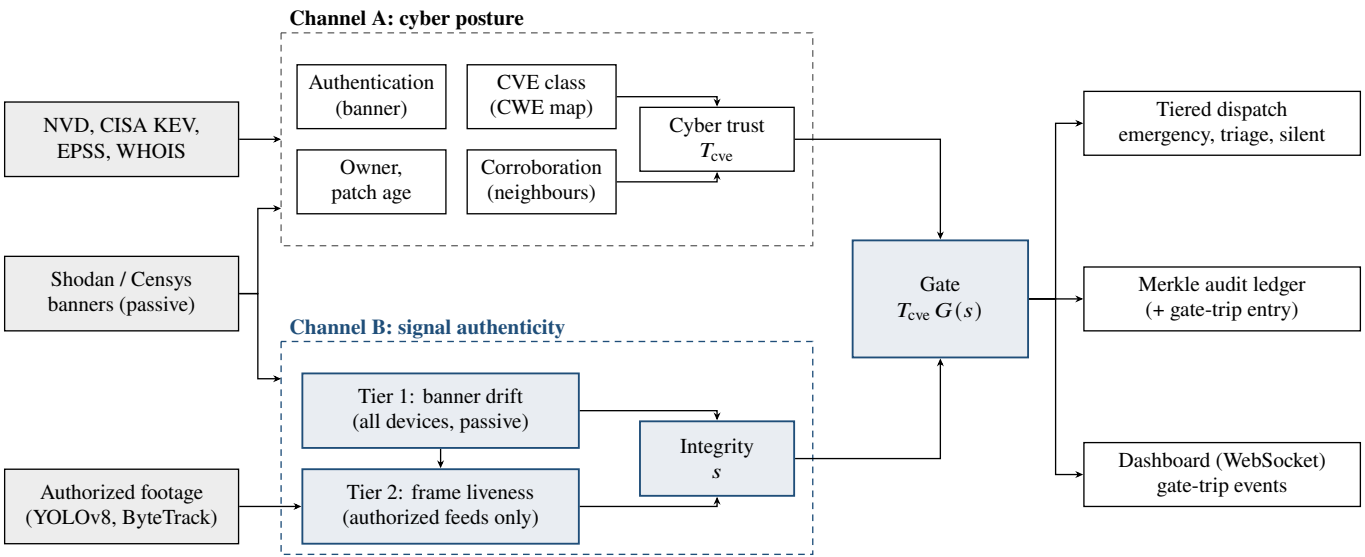

Channel A: cyber posture
NVD, CISA KEV, EPSS, WHOIS
Authentication (banner)
CVE class (CWE map)
Cyber trust T_{cve}
Owner, patch age
Corroboration (neighbours)
Shodan / Censys banners (passive)
Channel B: signal authenticity
Tier 1: banner drift (all devices, passive)
Integrity s
Authorized footage (YOLOv8, ByteTrack)
Tier 2: frame liveness (authorized feeds only)
Gate $T_{\text{cve}}\, G(s)$
Tiered dispatch emergency, triage, silent
Merkle audit ledger (+ gate-trip entry)
Dashboard (WebSocket) gate-trip events